

CHUQUR O'RGANISH USULLARI · Ma'ruza konspekti (Lecture Note)

# 4-ma'ruza. Rekurrent neyron tarmoqlar: RNN, LSTM va GRU

2026–2027 o'quv yili

**Reja.** 1) Ketma-ketlik ma'lumotlari va masala turlari. 2) Oddiy RNN (Elman tarmog'i). 3) BPTT va gradientning so'nishi/portlashi. 4) LSTM: darvozalar va yacheyka holati. 5) GRU. 6) Chuqur, ikki yo'nalishli RNN, seq2seq, til modeli va perplexity. 7) Tajriba natijalari va saboqlar. 8) Python (NumPy, PyTorch). 9) Nazorat savollari va mashqlar.

**O'quv natijalari.** Ma'ruzadan so'ng talaba: RNN, LSTM va GRU tenglamalarini yozadi va bitta qadamni qo'lda hisoblaydi; BPTT formulasini keltirib chiqaradi va gradient so'nishini yakobianlar ko'paytmasi orqali isbotlaydi; LSTM ning qo'shuvchi xotira yo'li nega yordam berishini tushuntiradi; parametrlar sonini hisoblaydi; gradient kesish, unutish darvozasi siljishi va bazaviy usul bilan taqqoslashni amalda qo'llaydi.

**Tayanch iboralar.** ketma-ketlik, yashirin holat (hidden state), RNN, parametrlarni bo'lishish, BPTT, truncated BPTT, yakobian, spektral radius, vanishing/exploding gradient, gradient clipping, LSTM, yacheyka holati, forget/input/output gate, GRU, update/reset gate, bidirectional RNN, seq2seq, teacher forcing, perplexity.

## 1. Ketma-ketlik ma'lumotlari va masala turlari

Matn, nutq, vaqt qatorlari (harorat, narx, sensor signallari), DNK — bularning barchasi *ketma-ketlik*:  $x_{1:T} = (x_1, \dots, x_T)$ ,  $x_t \in \mathbb{R}^d$ , bunda uzunlik  $T$  namunadan namunaga o'zgaradi va elementlar *tartibi* ma'noga ega. To'liq bog'langan tarmoq (MLP) qat'iy o'lchamli kirish talab qiladi. Ikki sodda yo'l ham yaroqsiz:

- *barchasini bitta vektorga ulash* — parametrlar soni  $T_{\max}$  ga proporsional,  $t$ - va  $(t+1)$ -pozitsiyadagi bir xil so'z butunlay boshqa og'irliklar bilan ishlanadi;
- *o'rtachalash* — tartib yo'qoladi: “it odamni qopdi” va “odam itni qopdi” bir xil vektorga aylanadi.

Yechim — vaqt bo'ylab *bitta* funksiyani qayta-qayta qo'llash va o'tmish haqidagi axborotni *yashirin holat*  $h_t$  da saqlash. Bu CNN dagi og'irliklarni bo'lishish g'oyasining vaqt o'qidagi analogi.

**Masala turlari.** *many-to-one* (matn sentiment: oxirgi  $h_T$  dan sinf); *many-to-many, sinxron* (so'z turkumlarini belgilash: har bir  $h_t$  dan  $y_t$ ); *many-to-many, asinxron* (mashina tarjimasini, seq2seq: uzunliklar farq qiladi); *one-to-many* (tasvirga izoh yozish).

**Kirishni ifodalash.** Diskret tokenlar (so'z, belgi, subword) lug'atdagi indeks  $w_t \in \{1, \dots, V\}$  bilan beriladi. *One-hot* vektor  $e_{w_t} \in \mathbb{R}^V$  siyrak va o'xshashlikni aks ettirmaydi; shuning uchun o'rgatiladigan *embedding* matritsasi  $E \in \mathbb{R}^{V \times d}$  ishlatiladi:  $x_t = E^T e_{w_t}$  ( $E$  ning  $w_t$ -qatori). Uzlüksiz signallar (vaqt qatorlari) odatda standartlashtirilib ( $\mu = 0$ ,  $\sigma = 1$ ) to'g'ridan-to'g'ri beriladi.

**Paketlash.** Turli uzunlikdagi ketma-ketliklar paketga eng uzuni bo'yicha nollar bilan to'ldiriladi (padding). Masalan, uzunliklari 5, 2, 7 bo'lgan,  $d = 3$  o'lchamli uchta ketma-ketlik  $3 \times 7 \times 3$  tenzorga aylanadi; 21 pozitsiyadan faqat 14 tasi haqiqiy. To'ldiruvchi nollar ham RNN dan o'tsa, oxirgi holat buziladi; PyTorch'da `pack_padded_sequence` har ketma-ketlikni o'z uzunligida to'xtatadi, yo'qotishda esa to'ldirilgan pozitsiyalar niqob bilan chiqarib tashlanadi.

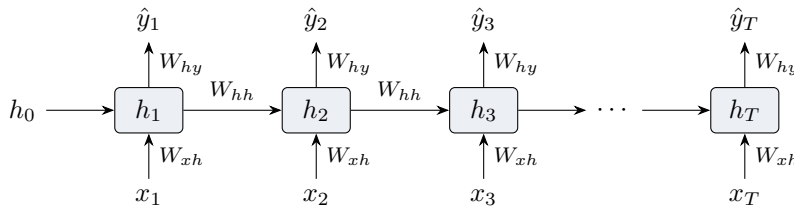
## 2. Oddiy rekurrent tarmoq (RNN)

**Ta'rif 2.1** (Elman RNN). Parametrlar  $W_{xh} \in \mathbb{R}^{H \times d}$ ,  $W_{hh} \in \mathbb{R}^{H \times H}$ ,  $W_{hy} \in \mathbb{R}^{m \times H}$ ,  $b_h \in \mathbb{R}^H$ ,  $b_y \in \mathbb{R}^m$  bo'lsin.  $h_0 = \mathbf{0}$  va  $t = 1, \dots, T$  uchun

$$a_t = W_{xh}x_t + W_{hh}h_{t-1} + b_h, \quad h_t = \tanh(a_t), \quad o_t = W_{hy}h_t + b_y, \quad \hat{y}_t = \text{softmax}(o_t).$$

Barcha qadamlarda *bir xil* parametrlar ishlatiladi (parametrlarni bo'lishish).

$W_{xh}x_t$  — yangi axborot,  $W_{hh}h_{t-1}$  — xotira;  $\tanh$  natijani  $(-1, 1)$  ga siqadi. Hisoblash grafini vaqt bo'yicha “yoyish” (unrolling) chuqurligi  $T$  ga teng, lekin og'irliklari umumiy bo'lgan oddiy to'g'ri tarqaluvchi tarmoqni beradi (1-rasm).



1-rasm. RNN ning vaqt bo'yicha yoyilgan grafi: har bir strelkadagi matritsa barcha qadamlarda bir xil.

**Xossa 2.2** (Parametrlar soni). Bitta RNN qatlami  $H(d + H) + H$  ta parametrga ega (chiqish qatlami-siz). PyTorch `nn.RNN` ikkita siljish vektori ( $b_{ih}$ ,  $b_{hh}$ ) saqlaydi, shuning uchun unda  $H(d + H) + 2H$ . Parametrlar soni  $T$  ga *bog'liq emas*.

**Misol 2.3.** Skalyar RNN:  $w_{xh} = 1$ ,  $w_{hh} = 0,5$ ,  $b = 0$ , kirish  $x = (1; 0; 0)$  (signal faqat birinchi qadamda).  $h_1, h_2, h_3$  ni toping.

*Yechim.*  $h_1 = \tanh(1) = 0,7616$ ;  $h_2 = \tanh(0,5 \cdot 0,7616) = \tanh(0,3808) = 0,3634$ ;  $h_3 = \tanh(0,1817) = 0,1797$ . Signal har qadamda taxminan ikki marta kamaymoqda: kichik argumentda  $\tanh u \approx u$ , demak  $h_t \approx 0,5^{t-1} h_1$ .  $\triangleleft$

### 3. Vaqt bo'yicha orqaga tarqalish (BPTT) va gradient muammolari

Ketma-ketlik bo'yicha yo'qotish  $L = \sum_{t=1}^T L_t(\hat{\mathbf{y}}_t, \mathbf{y}_t)$ . Yoyilgan graf oddiy chuqur tarmoq bo'lgani uchun unga orqaga tarqalish qo'llanadi — bu *BPTT* (backpropagation through time) deyiladi. Farqi faqat shundaki, umumiy parametr bo'yicha gradient barcha qadamlardagi hissalar yig'indisi bo'ladi.

**Teorema 3.1** (BPTT formulalari).  $\delta_t := \partial L / \partial \mathbf{h}_t$  (to'liq hosila) belgilansin. U holda  $\delta_T = W_{hy}^T(\hat{\mathbf{y}}_T - \mathbf{y}_T)$  va  $t = T - 1, \dots, 1$  uchun

$$\delta_t = W_{hy}^T(\hat{\mathbf{y}}_t - \mathbf{y}_t) + W_{hh}^T[(1 - \mathbf{h}_{t+1}^2) \odot \delta_{t+1}],$$

$$\frac{\partial L}{\partial W_{hh}} = \sum_{t=1}^T [(1 - \mathbf{h}_t^2) \odot \delta_t] \mathbf{h}_{t-1}^T, \quad \frac{\partial L}{\partial W_{xh}} = \sum_{t=1}^T [(1 - \mathbf{h}_t^2) \odot \delta_t] \mathbf{x}_t^T.$$

(Softmax + cross-entropy uchun  $\partial L_t / \partial \mathbf{o}_t = \hat{\mathbf{y}}_t - \mathbf{y}_t$ .)

*Isbot.*  $\mathbf{h}_t$  yo'qotishga ikki yo'l bilan ta'sir qiladi:  $\mathbf{o}_t$  orqali (bevosita) va  $\mathbf{h}_{t+1} = \tanh(\mathbf{a}_{t+1})$  orqali (keyingi qadamlar). Zanjir qoidasi:  $\partial \mathbf{a}_{t+1} / \partial \mathbf{h}_t = W_{hh}$ ,  $\partial \mathbf{h}_{t+1} / \partial \mathbf{a}_{t+1} = \text{diag}(1 - \mathbf{h}_{t+1}^2)$ , bundan birinchi formula.  $W_{hh}$  har bir  $\mathbf{a}_t$  ga  $W_{hh} \mathbf{h}_{t-1}$  orqali kiradi;  $\partial L / \partial \mathbf{a}_t = (1 - \mathbf{h}_t^2) \odot \delta_t$  ni  $\mathbf{h}_{t-1}^T$  ga tashqi ko'paytirib,  $t$  bo'yicha yig'amiz.  $\square$

**Natija 3.2** (Yakobianlar ko'paytmasi).  $\delta_t$  rekursiyasini yoyib chiqsak,  $T$ -qadamdagi hissa uzoq o'tmishga

$$\frac{\partial \mathbf{h}_T}{\partial \mathbf{h}_t} = \prod_{k=t+1}^T \frac{\partial \mathbf{h}_k}{\partial \mathbf{h}_{k-1}} = D_T W_{hh} D_{T-1} W_{hh} \cdots D_{t+1} W_{hh}, \quad D_k = \text{diag}(1 - \mathbf{h}_k^2)$$

orqali yetadi (ko'paytuvchilar chapdan o'ngga  $k = T, \dots, t + 1$  tartibida).

**Teorema 3.3** (Gradient so'nishining yetarli sharti).  $\gamma = \sup |\varphi'|$  ( $\tanh$  uchun  $\gamma = 1$ ) va  $s = \|W_{hh}\|_2$  (eng katta singulyar son) bo'lsin. U holda

$$\left\| \frac{\partial \mathbf{h}_T}{\partial \mathbf{h}_t} \right\|_2 \leq (\gamma s)^{T-t}.$$

Demak  $\gamma s < 1$  bo'lsa, uzoq bog'lanishlar hissasi  $T - t$  bo'yicha eksponensial so'nadi. Aksincha, gradient portlashi uchun  $\rho(W_{hh}) > 1/\gamma$  zarur (lekin yetarli emas).

*Isbot.* Normaning submultiplikativligi:  $\|D_k W_{hh}\|_2 \leq \|D_k\|_2 \|W_{hh}\|_2 \leq \gamma s$ ;  $T - t$  ta ko'paytuvchi olamiz. Zaruriylik qismi — Pascanu va boshq. (2013) natijasi:  $\rho < 1/\gamma$  bo'lsa, uzun ko'paytmalar chegaralangan va nolga intiladi.  $\square$

**Izoh 3.4.** Adabiyotda ba'zan yakobian  $\prod \text{diag}(1 - h_k^2) W_{hh}^T$  ko'rinishida yoziladi. Yakobian  $\partial \mathbf{h}_k / \partial \mathbf{h}_{k-1} = D_k W_{hh}$  (transponirlanmagan);  $W_{hh}^T$  esa gradient *vektori* orqaga uzatilganda paydo bo'ladi (3.1-teorema). Normalar bir xil, shuning uchun xulosa o'zgarmaydi.

**Misol 3.5.** 2.3-misoldagi RNN uchun  $\partial h_3 / \partial h_1$  va  $\partial h_3 / \partial x_1$  ni toping.

*Yechim.*  $\partial h_3 / \partial h_1 = (1 - h_3^2) w_{hh} (1 - h_2^2) w_{hh} = (0,9677 \cdot 0,5)(0,8679 \cdot 0,5) = 0,2100$ ;  $\partial h_3 / \partial x_1 = 0,2100 \cdot (1 - h_1^2) w_{xh} = 0,2100 \cdot 0,4200 = 0,0882$ . Ikki qadamda sezuvchanlik  $\approx 5$  marta,  $\partial h_3 / \partial x_1$  esa 11 marta kichraydi.  $\triangleleft$

**Misol 3.6** (BPTT to'liq hisobi). 2.3-misoldagi tarmoqda faqat oxirgi qadamda yo'qotish bor:  $L = \frac{1}{2}(h_3 - y)^2$ ,  $y = 0,5$ .  $\partial L / \partial w_{hh}$  va  $\partial L / \partial w_{xh}$  ni toping.

*Yechim.*  $L = \frac{1}{2}(0,1797 - 0,5)^2 = 0,05129$ . Orqaga yurish ( $g_t := \partial L / \partial a_t = \delta_t(1 - h_t^2)$ ):

| $t$ | $\delta_t = \partial L / \partial h_t$ | $1 - h_t^2$ | $g_t$    | $g_t h_{t-1}$   | $g_t x_t$ |
|-----|----------------------------------------|-------------|----------|-----------------|-----------|
| 3   | $h_3 - y = -0,32027$                   | 0,96770     | -0,30993 | -0,11263        | 0         |
| 2   | $w_{hh} g_3 = -0,15496$                | 0,86794     | -0,13450 | -0,10243        | 0         |
| 1   | $w_{hh} g_2 = -0,06725$                | 0,41997     | -0,02824 | 0 ( $h_0 = 0$ ) | -0,02824  |

$\partial L / \partial w_{hh} = \sum_t g_t h_{t-1} = -0,21506$ ,  $\partial L / \partial w_{xh} = \sum_t g_t x_t = -0,02824$ . Markaziy ayirma ( $\varepsilon = 10^{-6}$ ) ham  $-0,21506$  beradi.  $x_1$  ga yetib kelgan signal ( $g_1$ )  $g_3$  dan 11 marta kichik — so'nishning kichik ko'rinishi.  $\triangleleft$

**Tajriba (kitob).**  $W_{hh}$  ning spektral radiusi  $\rho$  aniq berilgan qiymatga keltirilib, 60 qadamdan keyin  $\|\partial \mathbf{h}_T / \partial \mathbf{h}_0\|$  o'lchangan:

| $\rho$ | 0,5                   | 0,9                  | 1,0                  | 1,5                  |
|--------|-----------------------|----------------------|----------------------|----------------------|
| norma  | $1,25 \cdot 10^{-19}$ | $1,34 \cdot 10^{-4}$ | $4,15 \cdot 10^{-2}$ | $3,44 \cdot 10^{-3}$ |

$\rho = 1,5$  da portlash kuzatilmadi: katta  $W_{hh}$  da  $\mathbf{h}_k \rightarrow \pm 1$  (to'yinish),  $D_k \approx 0$  va bu ko'paytuvchi  $W_{hh}$  ning kuchaytirishini bosib ketadi.  $\rho > 1$  — portlash uchun faqat zarur shart. Amalda tanh-RNN da asosiy muammo — *so'nish*.

**Boshlang'ich qiymat.** So'nish/portlashni kamaytirish uchun  $W_{hh}$  ortogonal matritsa bilan boshlanadi ( $\|W_{hh}\|_2 = 1$ , barcha singulyar sonlar 1); ReLU-RNN uchun  $W_{hh} = I$ ,  $\mathbf{b} = \mathbf{0}$  (IRNN, Le va boshq., 2015).  $W_{xh}$  uchun Xavier (Glorot) usuli ishlatiladi.

**Ta'rif 3.7** (Gradient kesish (gradient clipping by norm)). Barcha parametrlar gradienti  $\mathbf{g}$  uchun, chegara  $\tau > 0$ :  $\mathbf{g} \leftarrow \mathbf{g} \cdot \min(1, \tau / \|\mathbf{g}\|_2)$ . Yo'nalish saqlanadi, faqat uzunlik cheklanadi.

**Misol 3.8.**  $\mathbf{g} = (3; 4)$ ,  $\tau = 1$ :  $\|\mathbf{g}\| = 5$ ,  $\mathbf{g} \leftarrow (0,6; 0,8)$ . Kitobdagi tajribada  $\rho \approx 1,8$  bo'lgan RNN da 60 qadamdan keyin norma 28,92 ga yetgan, kesishdan keyin aynan 1,00. PyTorch'dagi `clip_grad_norm_kesishdan oldingi` normani qaytaradi.

**Truncated BPTT.** Juda uzun ketma-ketlik  $k$  uzunlikdagi bo'laklarga bo'linadi; yashirin holat bo'laklar orasida uzatiladi ( $\mathbf{h} = \mathbf{h}.\text{detach}()$ ), gradient esa faqat bo'lak ichida hisoblanadi. Xotira  $O(kH)$ , lekin  $k$  dan uzoq bog'lanishlar gradient orqali o'rganilmaydi.

#### Amaliy talqin: so'nish nimani anglatadi

Gradient so'nishi — model uzoq bog'lanishni “ko'ra olmaydi” degani emas (oldinga yurishda axborot  $\mathbf{h}_t$  da qolishi mumkin), balki *o'rgana olmaydi* degani:  $x_1$  ning  $L_T$  ga ta'sirini to'g'rilovchi signal yetib kelmaydi. Matnda “Men Toshkentda tug'ilganman ... (40 so'z) ... shuning uchun ona tilim — ...” kabi

bog'lanish aynan shu sababli oddiy RNN uchun qiyin.

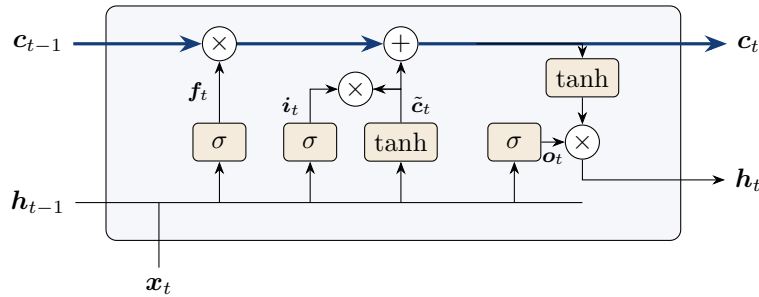
#### 4. LSTM: darvozali xotira

G'oya (Hochreiter, Schmidhuber, 1997): xotirani matritsaga *ko'paytirib* emas, *qo'shib* uzatish. Buning uchun alohida *yacheyka holati* (cell state)  $c_t$  kiritiladi.

**Ta'rif 4.1** (LSTM yacheykasi).  $z_t = [h_{t-1}; x_t] \in \mathbb{R}^{H+d}$ ,  $\sigma$  — sigmoid,  $\odot$  — Adamar ko'paytmasi:

$$\begin{aligned} f_t &= \sigma(W_f z_t + b_f) && (\text{forget gate} — \text{unutish darvozasi}), \\ i_t &= \sigma(W_i z_t + b_i) && (\text{input gate} — \text{kirish darvozasi}), \\ \tilde{c}_t &= \tanh(W_c z_t + b_c) && (\text{nomzod qiymat}), \\ o_t &= \sigma(W_o z_t + b_o) && (\text{output gate} — \text{chiqish darvozasi}), \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, && h_t = o_t \odot \tanh(c_t). \end{aligned}$$

Darvozalar  $(0, 1)$  dagi “ventillar”:  $f_t$  — eski xotiraning qancha qismi saqlanadi,  $i_t$  — yangi nomzodning qancha qismi yoziladi,  $o_t$  — xotiraning qaysi qismi hozir tashqariga chiqariladi.



2-rasm. LSTM yacheykasi: yuqoridagi qalin chiziq —  $c_t$  “konveyer lentasi”; unda faqat elementma-element  $\times$  va  $+$  amallari bor.

#### Algoritm (LSTM ning bitta qadami).

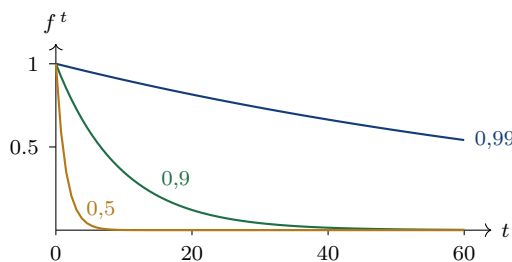
1.  $z \leftarrow [h_{t-1}; x_t]$ ;
2.  $f_t, i_t, o_t \leftarrow \sigma(\cdot)$ ,  $\tilde{c}_t \leftarrow \tanh(\cdot)$  (bitta  $4H \times (H + d)$  matritsa ko'paytmasi bilan);
3.  $c_t \leftarrow f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$  (*qo'shish*, matritsaga ko'paytirish yo'q);
4.  $h_t \leftarrow o_t \odot \tanh(c_t)$ ; ( $h_t, c_t$ ) qaytariladi.

**Xossa 4.2** (Qo'shuvchi xotira yo'li).  $c_t$  ning  $c_{t-1}$  ga *bevosita* bog'liqligi bo'yicha (darvozalar orqali bilvosita yo'llarsiz)  $\partial c_t / \partial c_{t-1} = \text{diag}(f_t)$ , demak bu yo'l bo'ylab

$$\left. \frac{\partial c_T}{\partial c_t} \right|_{\text{bevosita}} = \text{diag} \left( \prod_{k=t+1}^T f_k \right).$$

*Isbot.*  $c_k = f_k \odot c_{k-1} + i_k \odot \tilde{c}_k$  da  $f_k, i_k, \tilde{c}_k$  ni o'zgarmas deb qaraymiz; bu  $c_{k-1}$  ga nisbatan elementma-element chiziqli, koeffitsiyenti  $f_k$ . Zanjir qoidasi ko'paytmani beradi.  $\square$

Taqqoslang: RNN da har qadamda  $D_k W_{hh}$  (odatda normasi  $< 1$ ), LSTM da esa  $f_k \in (0, 1)$  ga ko'paytiriladi;  $f_k \approx 1$  bo'lsa,  $1^T = 1$  — axborot va gradient deyarli o'zgarmasdan o'tadi (*constant error carousel*). Masalan,  $f = 0,9$  da  $0,9^{50} = 5,15 \cdot 10^{-3}$ ,  $f = 0,99$  da  $0,99^{50} = 0,605$ .



3-rasm.  $c_t$  yo'li bo'ylab xotira/gradient ulushi  $f^t$ : kichik farq (0,9 va 0,99) uzoq ufqda katta farq beradi.

**Unutish darvozasi siljishi.** Tasodifiy boshlang'ich qiymatda  $b_f \approx \mathbf{0}$ , demak  $f_t \approx \sigma(0) = 0,5$  va xotira  $0,5^t$  bo'yicha so'nadi. Shuning uchun  $b_f$  ni  $+1...+2$  qilib boshlash keng tarqalgan (Jozefowicz va boshq., 2015):  $\sigma(1) = 0,731$ ,  $\sigma(2) = 0,881$ ,  $\sigma(4) = 0,982$ .

**Misol 4.3.** Skalyar LSTM ( $H = d = 1$ ):  $x_1 = 1$ ,  $h_0 = 0$ ,  $c_0 = 0,5$ ; og'irliklar shunday tanlangan-ki,  $a_f = x + 1$ ,  $a_i = 0,5x$ ,  $a_c = x$ ,  $a_o = x$  ( $h_0 = 0$  bo'lgani uchun  $h$  ga oid og'irliklar ahamiyatsiz).  $c_1$  va  $h_1$  ni toping.

*Yechim.*  $f_1 = \sigma(2) = 0,8808$ ;  $i_1 = \sigma(0,5) = 0,6225$ ;  $\tilde{c}_1 = \tanh 1 = 0,7616$ ;  $o_1 = \sigma(1) = 0,7311$ .  
 $c_1 = 0,8808 \cdot 0,5 + 0,6225 \cdot 0,7616 = 0,4404 + 0,4741 = 0,9145$ ;  $h_1 = 0,7311 \cdot \tanh(0,9145) = 0,7311 \cdot 0,7233 = 0,5287$ .  $\triangleleft$

## 5. GRU: soddalashtirilgan darvozali yacheyka

GRU (Cho va boshq., 2014) alohida  $c_t$  saqlamaydi va ikkita darvozadan foydalanadi. Quyida PyTorch konvensiyasi keltirilgan:

$$\begin{aligned} r_t &= \sigma(W_{ir}x_t + W_{hr}h_{t-1} + b_r) && \text{(reset gate),} \\ u_t &= \sigma(W_{iu}x_t + W_{hu}h_{t-1} + b_u) && \text{(update gate),} \\ \tilde{h}_t &= \tanh(W_{in}x_t + r_t \odot (W_{hn}h_{t-1} + b_{hn}) + b_{in}), \\ h_t &= (1 - u_t) \odot \tilde{h}_t + u_t \odot h_{t-1}. \end{aligned}$$

$h_t$  — eski holat va nomzodning *qavariq kombinatsiyasi*:  $u_t \rightarrow 1$  da  $h_t = h_{t-1}$  (xotira to'liq saqlanadi,  $\partial h_t / \partial h_{t-1} \approx I$ ). Ba'zi manbalarda (Cho, 2014)  $u_t$  va  $1 - u_t$  o'рни almashgan — bu faqat belgilash.

**Xossa 5.1.** GRU da  $\frac{\partial h_t}{\partial h_{t-1}} = \text{diag}(u_t) + \text{diag}(1 - u_t) \frac{\partial \tilde{h}_t}{\partial h_{t-1}} + \text{diag}(h_{t-1} - \tilde{h}_t) \frac{\partial u_t}{\partial h_{t-1}}$ . Birinchi had — LSTM dagi  $\text{diag}(f_t)$  analogi:  $u_t \approx \mathbf{1}$  da yakobian  $\approx I$  va gradient so'nmaydi.

*Isbot.*  $h_t = \tilde{h}_t + u_t \odot (h_{t-1} - \tilde{h}_t)$  ni  $h_{t-1}$  bo'yicha differensiallaymiz (ko'paytma qoidasi):  $u_t \odot (\cdot)$  dan  $\text{diag}(u_t)(I - \partial \tilde{h}_t / \partial h_{t-1})$  va  $\text{diag}(h_{t-1} - \tilde{h}_t) \partial u_t / \partial h_{t-1}$ ;  $\partial \tilde{h}_t / \partial h_{t-1}$  bilan yig'ib, talab qilingan ifodani olamiz.  $\square$

**Misol 5.2.** Skalyar GRU:  $x = 1$ ,  $h_0 = 0,5$ ,  $a_u = 0,5x$ ,  $a_r = x$ ,  $\tilde{h} = \tanh(x + r h_0)$ .  $h_1$  ni toping.

*Yechim.*  $u = \sigma(0,5) = 0,6225$ ,  $r = \sigma(1) = 0,7311$ ,  $\tilde{h} = \tanh(1 + 0,3655) = \tanh(1,3655) = 0,8777$ ;  
 $h_1 = 0,3775 \cdot 0,8777 + 0,6225 \cdot 0,5 = 0,3313 + 0,3113 = 0,6426$ .  $\triangleleft$

**Xossa 5.3** (Parametrlar nisbati). Bir xil  $d$ ,  $H$  da PyTorch hisobida: RNN —  $H(d + H) + 2H$ , GRU —  $3[H(d + H) + 2H]$ , LSTM —  $4[H(d + H) + 2H]$ ; nisbat RNN : GRU : LSTM = 1 : 3 : 4. Masalan,  $d = 10$ ,  $H = 20$  da 640, 1920, 2560.

|              | RNN                      | LSTM                            | GRU                  |
|--------------|--------------------------|---------------------------------|----------------------|
| Holat        | $h_t$                    | $h_t, c_t$                      | $h_t$                |
| Darvozalar   | —                        | 3 ( $f, i, o$ )                 | 2 ( $r, u$ )         |
| Xotira yo'li | $D_k W_{hh}$ ko'paytmasi | $f_t$ ga ko'paytma (qo'shuvchi) | $u_t$ interpolatsiya |
| Parametrlar  | $1 \times$               | $4 \times$                      | $3 \times$           |

## 6. Arxitektura variantlari va qo'llanishlar

**Chuqur (stacked) RNN:**  $\ell$ -qatlamning kirishi ( $\ell - 1$ )-qatlamning  $h_t^{(\ell-1)}$  ketma-ketligi. **Ikki yo'nalishli (bidirectional) RNN:**  $\vec{h}_t$  chapdan o'ngga,  $\overleftarrow{h}_t$  o'ngdan chapga hisoblanadi,  $h_t = [\vec{h}_t; \overleftarrow{h}_t]$ ; butun ketma-ketlik oldindan ma'lum bo'lsa (tasniflash, belgilash) foydali, lekin real vaqtli bashoratda ishlatib bo'lmaydi, chunki kelajakni talab qiladi.

**seq2seq (encoder-decoder).** Encoder kirishni  $\mathbf{c} = \mathbf{h}_T^{\text{enc}}$  vektorga siqadi, decoder undan chiqish ketma-ketligini token-ma-token hosil qiladi:  $p(y_{1:S} | x_{1:T}) = \prod_s p(y_s | y_{<s}, \mathbf{c})$ . O'qitishda decoder kirishiga oldingi *haqiqiy* token beriladi (*teacher forcing*); inferensiyada — modelning o'z bashorati (greedy yoki beam search). Bitta  $\mathbf{c}$  vektor “tor bo'g'iz” bo'lgani uchun Bahdanau va boshq. (2015) har bir decoder qadamida encoder holatlarining og'irlikli yig'indisini ishlatishni taklif qilgan — bu *attention* (keyingi ma'ruza).

**Ta'rif 6.1** (Bahdanau (additive) attention). Decoder holati  $\mathbf{s}_{s-1}$  va encoder holatlari  $\mathbf{h}_1, \dots, \mathbf{h}_T$  uchun

$$e_{s,t} = \mathbf{v}^T \tanh(W\mathbf{s}_{s-1} + U\mathbf{h}_t), \quad \alpha_{s,t} = \frac{\exp e_{s,t}}{\sum_{\tau} \exp e_{s,\tau}}, \quad \mathbf{c}_s = \sum_{t=1}^T \alpha_{s,t} \mathbf{h}_t.$$

$\alpha_{s,t} \geq 0, \sum_t \alpha_{s,t} = 1$  — “ $s$ -chiqish tokeni uchun  $t$ -kirish tokeni qanchalik muhim”.

**Misol 6.2.**  $T = 3$ , skalyar holatlar  $h = (1; 0; -1)$ , ballar  $e = (1; 2; 0)$ . U holda  $\alpha = \text{softmax}(e) = (0,2447; 0,6652; 0,0900)$ ,  $c = 0,2447 \cdot 1 + 0,6652 \cdot 0 + 0,0900 \cdot (-1) = 0,1547$ . Gradient  $\mathbf{c}_s$  dan har bir  $\mathbf{h}_t$  ga *bitta* qadamda o'tadi —  $T$  ta yakobian ko'paytmasi yo'q.

**Beam search (nur qidiruvi).** Greedy dekodlash har qadamda eng ehtimolli tokenni oladi va global optimumni o'tkazib yuborishi mumkin. Beam search ( $k$  — nur kengligi):

1.  $\mathcal{B} \leftarrow \{(\langle \text{bos} \rangle, 0)\}$  (gipoteza, log-ehitimol);
2. har bir gipotezani barcha  $V$  token bilan davom ettirib, ballni  $+\log p(y | \cdot)$  qilib yangilash;
3.  $kV$  nomzoddan eng yaxshi  $k$  tasini qoldirish;  $\langle \text{eos} \rangle$  bilan tugaganlarini alohida saqlash;
4. maksimal uzunlik yoki barcha nurlar tugaguncha 2–3 ni takrorlash; uzunlik bo'yicha normallashtirilgan eng yaxshi ballni qaytarish.

Murakkablik  $O(SkV)$ ;  $k = 1$  — greedy.

**Misol 6.3** (Greedy va beam search).  $p(y_1)$ :  $A$ : 0,5,  $B$ : 0,4,  $C$ : 0,1;  $p(y_2 | A)$ :  $A$ : 0,4,  $B$ : 0,3,  $C$ : 0,3;  $p(y_2 | B)$ :  $A$ : 0,9,  $B$ : 0,05,  $C$ : 0,05. Ikki tokenli eng ehtimolli ketma-ketlikni toping.

*Yechim.* Greedy:  $y_1 = A$ , so'ng  $y_2 = A$ , ehtimol  $0,5 \cdot 0,4 = 0,20$ . Beam ( $k = 2$ ): 1-qadamda  $\{A, B\}$  saqlanadi; nomzodlar  $AA$  0,20,  $AB$  0,15,  $AC$  0,15,  $BA$  0,36,  $BB$  0,02,  $BC$  0,02; eng yaxshisi  $BA$  (0,36). Greedy birinchi qadamdagi “ochko'zlik” tufayli yutqazdi.  $\triangleleft$

**Exposure bias.** Teacher forcing bilan o'qitilgan decoder inferensiyada o'z xatolari ustiga quradi — o'qitish va inferensiya taqsimotlari farq qiladi. Yumshatish usullari: scheduled sampling (haqiqiy token o'rniga ba'zan model bashoratini berish), beam search, ketma-ketlik darajasidagi yo'qotishlar.

**Misol 6.4** (So'z darajasidagi LSTM til modeli).  $V = 5\,000$ , embedding o'lchami  $E = 128$ ,  $H = 256$ , bitta LSTM qatlami, chiqish  $\text{Linear}(H, V)$ . Parametrlar: embedding  $5\,000 \cdot 128 = 640\,000$ ; LSTM  $4(256 \cdot 384 + 2 \cdot 256) = 395\,264$ ; chiqish  $256 \cdot 5\,000 + 5\,000 = 1\,285\,000$ ; jami  $2\,320\,264$ . Parametrlarning 83% i lug'at bilan bog'liq qatlamlarda — shuning uchun ko'pincha embedding va chiqish matritsasi bo'lishiladi (weight tying,  $E = H$  bo'lganda).

**Ta'rif 6.5** (Til modeli va perplexity). Til modeli  $p(w_{1:N}) = \prod_t p(w_t | w_{<t})$  ni baholaydi; RNN da  $p(w_t | w_{<t}) = \text{softmax}(W_{hy}\mathbf{h}_{t-1} + \mathbf{b}_y)$ . Test matnida

$$\text{PPL} = \exp\left(-\frac{1}{N} \sum_{t=1}^N \log p(w_t | w_{<t})\right) = \exp(\text{o'rtacha cross-entropy}).$$

PPL — model har qadamda “nechta teng ehtimolli variant orasida ikkilanayotganini” bildiradi:  $V$  ta belgi uchun tekis taqsimotda  $\text{PPL} = V$ .

**Misol 6.6.** To'g'ri tokenlarga model 0,5; 0,25; 0,125 ehtimol bergan. O'rtacha  $\text{CE} = (\ln 2 + \ln 4 + \ln 8)/3 = 2 \ln 2 = 1,386$  nat,  $\text{PPL} = e^{2 \ln 2} = 4$ .

## 7. Tajriba natijalari va saboqlar

**Gradientning yetib borish masofasi.** O'qitilmagan yacheyka  $T = 120$  qadam ishlatiladi va  $\text{reach}(t) = \|\partial(\sum \mathbf{h}_T)/\partial \mathbf{x}_t\|$  o'lchanadi; *samarali ufq* —  $\text{reach}(t)$  maksimumining  $10^{-3}$  qismidan katta bo'lib qoladigan qadamlar soni. Bu — optimallashtirish omadiga bog'liq bo'lmagan arxitektura xossasi.

| Yacheyka             | RNN | GRU | LSTM | LSTM, $b_f = +2$ | LSTM, $b_f = +4$ |
|----------------------|-----|-----|------|------------------|------------------|
| Samarali ufq (qadam) | 9   | 14  | 16   | 120              | 120              |

$b_f = 0$  da LSTM ning ustunligi kichik ( $f \approx 0,5$ );  $b_f = +2$  da ( $f \approx 0,88$ ) ufq butun ketma-ketlikka yetadi. LSTM afzalligi darvozalarining *ochiq* bo'lishiga bog'liq.

**Eslab qolish masalasi.** Birinchi qadamda  $\pm 1$  ishora, keyin faqat  $\mathcal{N}(0, 1)$  shovqin; oxirida ishorani aytish kerak. 4 ta urug' (seed) bilan:  $T = 10$  da hamma model 4/4;  $T = 40$  da RNN 3/4, GRU 1/4, LSTM 0/4;  $T = 80$  da RNN 1/4 (o'rtacha aniqlik 0,60), GRU 1/4, LSTM 0/4. Bu nazariyaga zid ko'rinadi, lekin sabab *optimallashtirishda*: LSTM da parametrlar 4 marta ko'p, boshlang'ich  $f \approx 0,5$  va 40 epoxa yetmagan. Qobiliyat va uni topish — boshqa-boshqa narsalar; bitta urug'ga tayangan xulosa ishonchsiz.

**Vaqt qatori bashorati.** Ikki sinus yig'indisini bir qadam oldinga bashorat qilish (40 qadamli oyna): bazaviy usul "oxirgi qiymatni takrorla" MSE = 0,00566, LSTM MSE  $\approx 0,00001$ . Absolyut MSE o'z-o'zidan ma'nosiz — faqat bazaviy usul bilan taqqoslash ma'no beradi.

**Sirpanuvchi oyna (sliding window).** Qator  $s_1, \dots, s_N$  dan o'quv juftliklari  $((s_i, \dots, s_{i+W-1}), s_{i+W})$  tuziladi; ular vaqt bo'yicha (aralashtirmasdan) o'quv va test qismlarga bo'linadi, aks holda kelajak axboroti o'qitishga sizib kiradi.  $k$  qadam oldinga bashoratda model o'z chiqishini qayta kirishga beradi (rekursiv strategiya) va xato to'planadi.

### Amaliy talqin: amaliy retsept

1) Standart tanlov — LSTM yoki GRU, nn.RNN emas. 2)  $b_f = +1\dots+2$ . 3) Gradient kesish  $\tau \in [0,5; 5]$ . 4) Turli uzunlikdagi paketlar uchun `pack_padded_sequence` (aks holda to'ldiruvchi nollar holatni buzadi). 5) Paketlar orasida `h.detach()`. 6) Natijani bazaviy usul va bir necha urug' bilan taqqoslang. Uzun kontekstli zamonaviy NLP masalalarida RNN o'rnini Transformer egallagan, lekin oqimli (streaming), kam resursli va vaqt qatori masalalarida RNN ning  $O(1)$  xotirali qadami hamon qulay.

## 8. Python'da amalga oshirish

Quyidagi dastur NumPy'da RNN va BPTT ni yozadi,  $\partial L/\partial W_{hh}$  ning bitta elementini markaziy ayirma bilan tekshiradi, so'ng PyTorch'da parametrlar sonini va gradient kesishni ko'rsatadi.

```
import numpy as np, torch, torch.nn as nn
rng = np.random.default_rng(0)
d, H, T = 3, 4, 6
Wx, Wh = rng.normal(0, .5, (H, d)), rng.normal(0, .5, (H, H))
b, X = np.zeros(H), rng.normal(size=(T, d))
def forward(Wh):
    h, hs = np.zeros(H), [np.zeros(H)]
    for t in range(T):
        h = np.tanh(Wx @ X[t] + Wh @ h + b); hs.append(h)
    return hs
hs = forward(Wh) # L = sum(h_T)
dWh, dh = np.zeros_like(Wh), np.ones(H) # BPTT: dL/dh_T = 1
for t in range(T, 0, -1):
    da = dh * (1 - hs[t]**2) # tanh hosilasi
    dWh += np.outer(da, hs[t-1]); dh = Wh.T @ da
E = np.zeros((H, H)); E[1, 1] = 1e-6
num = (forward(Wh + E)[-1].sum() - forward(Wh - E)[-1].sum()) / 2e-6
print(f"BPTT: {dWh[1,1]:.6f} sonli: {num:.6f}")
for cell in (nn.RNN, nn.GRU, nn.LSTM):
    m = cell(10, 20); print(cell.__name__, sum(p.numel() for p in m.parameters()))
```

```
g = torch.tensor([3., 4.]); g.grad = torch.tensor([3., 4.])
print(torch.nn.utils.clip_grad_norm_([g], 1.0).item(), [round(v, 4) for v in g.grad.tolist()])
```

Natija:

```
BPTT: -0.241862 sonli: -0.241862
RNN 640
GRU 1920
LSTM 2560
5.0 [0.6, 0.8]
```

PyTorch'da LSTM siljishlari  $[i, f, g, o]$  tartibida saqlanadi; unutish darvozasi siljishini o'rnatish: `p.data[H:2*H].fill_(1.0)` (`bias_ih` va `bias_hh` ikkalasiga qo'llansa, samarali siljish 2,0 bo'ladi).

Ikkinchi dastur — LSTM bilan vaqt qatorini bir qadam oldinga bashorat qilish (many-to-one) va "oxirgi qiymatni takrorla" bazaviy usuli bilan taqqoslash (CPU da  $\approx 40$  s):

```
import torch, torch.nn as nn
torch.manual_seed(0)
t = torch.arange(0, 400, 0.1)
s = torch.sin(t) + 0.5 * torch.sin(0.3 * t) # vaqt qatori
W = 40
X = torch.stack([s[i:i+W] for i in range(len(s) - W)]).unsqueeze(-1)
y = s[W:].unsqueeze(-1)
Xtr, ytr, Xte, yte = X[:3000], y[:3000], X[3000:], y[3000:]
class Net(nn.Module):
    def __init__(self, H=32):
        super().__init__(); self.lstm = nn.LSTM(1, H, batch_first=True); self.out = nn.Linear(H, 1)
    def forward(self, x):
        h, _ = self.lstm(x); return self.out(h[:, -1]) # many-to-one
net = Net(); opt = torch.optim.Adam(net.parameters(), lr=1e-2)
for ep in range(30):
    for i in torch.randperm(3000).split(100):
        loss = ((net(Xtr[i]) - ytr[i])**2).mean()
        opt.zero_grad(); loss.backward()
        nn.utils.clip_grad_norm_(net.parameters(), 1.0); opt.step()
with torch.no_grad():
    mse = ((net(Xte) - yte)**2).mean().item()
naive = ((Xte[:, -1] - yte)**2).mean().item() # "oxirgi qiymat"
print(f"naive_MSE={naive:.2e} LSTM_MSE={mse:.2e}")
```

Natija (PyTorch 2.x, CPU; oxirgi raqamlar muhitga qarab biroz farq qilishi mumkin):

```
naive MSE=5.17e-03 LSTM MSE=1.14e-05
```

LSTM bazaviy usuldan taxminan 450 marta aniq; kitobdagi 0,00566 va 0,00001 natijalari bilan bir tartibda.

## 9. Xulosa

1. RNN yashirin holat  $\mathbf{h}_t = \tanh(W_{xh}\mathbf{x}_t + W_{hh}\mathbf{h}_{t-1} + \mathbf{b})$  orqali ixtiyoriy uzunlikdagi ketma-ketlikni bir xil parametrlar bilan qayta ishlaydi.
2. BPTT — yoyilgan grafdagi orqaga tarqalish; umumiy parametr gradienti qadamlar bo'yicha yig'indi.
3.  $\partial \mathbf{h}_T / \partial \mathbf{h}_t = \prod D_k W_{hh}$ ;  $\|\cdot\| \leq (\gamma \|W_{hh}\|_2)^{T-t}$  — gradient so'nishi; portlash uchun  $\rho > 1$  zarur, lekin yetarli emas; portlashga qarshi — gradient kesish.
4. LSTM xotirani qo'shuvchi yo'l  $\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t$  orqali uzatadi; bu yo'lga gradient faqat  $\mathbf{f}_t$  ga ko'paytiriladi;  $b_f > 0$  ufqni keskin uzaytiradi.
5. GRU — 2 darvozali, 3 barobar parametrlri yengil variant;  $\mathbf{h}_t$  — eski va yangi holat interpolyatsiyasi.
6. Til modeli sifati perplexity = exp(CE) bilan o'lchanadi; seq2seq ning tor bo'g'izi attention'ga olib keldi.
7. Qobiliyat  $\neq$  o'qitilganlik: natijalarni bir necha urug' va bazaviy usul bilan tekshirish shart.

## 10. Nazorat savollari

1. Nega MLP uzunligi o'zgaruvchan ketma-ketlik uchun noqulay?
2. RNN da parametrlarni bo'lishish nimani beradi?
3. Many-to-one va seq2seq masalalariga misol keltiring.
4. BPTT oddiy backpropagation'dan nimasi bilan farq qiladi?
5.  $\partial \mathbf{h}_T / \partial \mathbf{h}_t$  formulasini yozing.
6. Gradient so'nishining yetarli shartini isbotlang.
7. Nega  $\rho = 1,5$  da gradient portlamadi?
8. Gradient kesish yo'nalishini o'zgartiradimi?
9. Truncated BPTT qanday cheklov kiritadi?
10. LSTM ning uch darvozasi vazifasini ayting.
11. Nega  $\partial \mathbf{c}_t / \partial \mathbf{c}_{t-1} = \text{diag}(\mathbf{f}_t)$  muhim?
12. Unutish darvozasi siljishi nima uchun musbat qilinadi?
13. GRU ning LSTM dan farqlari qanday?
14. RNN, GRU, LSTM parametrlari nisbati qanday?
15. Bidirectional RNN ni qachon ishlatib bo'lmaydi?
16. Teacher forcing nima?
17. Perplexity ta'rifi va talqini qanday?
18. Nega bazaviy usul bilan taqqoslash shart?

## 11. Mashqlar (javoblari bilan)

### A guruh.

1. 2.3-misolni  $w_{hh} = 0,95$  uchun takrorlang.  $h_t$  necha qadamda 0,1 dan pastga tushadi? *Javob:*  $h_{26} = 0,1012$ ,  $h_{27} = 0,0958$  — 27-qadamda.
2.  $f_t \equiv 0,9$  va  $f_t \equiv 0,99$  da  $T = 50$  uchun  $\partial c_T / \partial c_0$  (bevosita yo'l). *Javob:*  $0,9^{50} = 5,15 \cdot 10^{-3}$ ;  $0,99^{50} = 0,605$ .
3.  $d = 10$ ,  $H = 20$ : RNN, GRU, LSTM parametrlari (bitta siljish vektori bilan). *Javob:* 620; 1 860; 2 480 (PyTorch'da: 640; 1 920; 2 560).
4.  $\mathbf{g} = (6; 8)$ ,  $\tau = 2$  bo'yicha kesing. *Javob:* (1,2; 1,6).
5. 29 belgisi lug'atda tekis taqsimotli modelning perplexity'si va CE si. *Javob:* PPL = 29, CE =  $\ln 29 = 3,367$  nat.
6.  $b_f = 2$  va  $b_f = 4$  da  $(W_f \mathbf{z} = 0)$   $f_t$  qiymati. *Javob:* 0,881; 0,982.

### B guruh.

7. Skalyar LSTM:  $c_0 = 1$ ,  $a_f = 1$ ,  $a_i = 0$ ,  $a_c = 0,5$ ,  $a_o = 0,5$ .  $c_1, h_1$  ni toping. *Javob:*  $c_1 = 0,7311 + 0,5 \cdot 0,4621 = 0,9621$ ;  $h_1 = 0,6225 \cdot \tanh(0,9621) = 0,4639$ .
8.  $f^{100} \geq 0,5$  bo'lishi uchun  $f$  va  $b_f$  (boshqa hadlar nol) qanday bo'lishi kerak? *Javob:*  $f \geq 0,5^{0,01} = 0,99309$ ,  $b_f \geq \ln \frac{f}{1-f} = 4,97$ .
9. 3.1-teoremadan  $\partial L / \partial W_{hy}$  va  $\partial L / \partial \mathbf{b}_h$  ni yozing. *Javob:*  $\sum_t (\hat{\mathbf{y}}_t - \mathbf{y}_t) \mathbf{h}_t^T$ ;  $\sum_t (1 - \mathbf{h}_t^2) \odot \delta_t$ .
10. GRU da  $\mathbf{u}_t \equiv \mathbf{1}$  va  $\mathbf{u}_t \equiv \mathbf{0}$  bo'lsa, model nimaga aylanadi? *Javob:*  $\mathbf{h}_t = \mathbf{h}_0$  (xotira muzlaydi);  $\mathbf{r}_t \equiv \mathbf{1}$  bilan esa oddiy RNN ga.
11.  $V = 10\,000$  lug'atli model test matnida o'rtacha CE = 4,6 nat berdi. PPL va bir tokeniga bit? *Javob:* PPL =  $e^{4,6} \approx 99,5$ ;  $4,6 / \ln 2 = 6,64$  bit.
12. Nega bidirectional LSTM keyingi so'zni bashorat qiluvchi til modeli sifatida ishlatilmaydi? *Javob:*  $\overleftarrow{\mathbf{h}}_t$  kelajakdagi (bashorat qilinishi kerak bo'lgan) tokenlarni ko'radi — axborot sizishi.
13. 3.6-misolda  $y = 0$  bo'lsa,  $\partial L / \partial w_{hh}$  ishorasi qanday va nega? *Javob:*  $\delta_3 = h_3 > 0$ , barcha  $g_t > 0$ , gradient musbat:  $w_{hh}$  ni kamaytirish  $h_3$  ni 0 ga yaqinlashtiradi.

### C guruh.

13. Chiziqli RNN  $\mathbf{h}_t = W \mathbf{h}_{t-1}$ ,  $W$  diagonallashadigan bo'lsin.  $\partial \mathbf{h}_T / \partial \mathbf{h}_0 = W^T$  ning xos qiymatlari orqali so'nish/portlash shartini keltiring. *Ko'rsatma:*  $W = P \Lambda P^{-1}$ ,  $W^T = P \Lambda^T P^{-1}$ ;  $|\lambda_i| < 1$  yo'nalishlar so'nadi,  $|\lambda_i| > 1$  — portlaydi.
14.  $W$  ortogonal va  $\varphi$  ayniy bo'lsa,  $\|\partial \mathbf{h}_T / \partial \mathbf{h}_t\|_2 = 1$  ekanini isbotlang (unitary/orthogonal RNN g'oyasi). *Ko'rsatma:* ortogonal matritsalar ko'paytmasi ortogonal.
15. 4.2-xossada bilvosita yo'llar (darvozalar  $\mathbf{h}_{t-1}$  orqali  $\mathbf{c}_{t-1}$  ga bog'liq) qanday qo'shimcha hadlar beradi? Ular nega odatda kichik? *Ko'rsatma:* har biri  $\sigma' \leq \frac{1}{4}$  va  $\sigma_{t-1}$ ,  $\tanh'$  ko'paytuvchilarini o'z ichiga oladi.
16. Eslab qolish masalasini LSTM bilan  $b_f = +2$  boshlang'ich qiymatda qayta o'tkazing (4 urug').  $T = 80$  natijasini kitob jadvali bilan solishtiring va xulosani asoslang.
17. Ikkinchi dasturda  $W = 40$  o'rniga  $W = 5$  va  $W = 100$  oling; sinus davri ( $2\pi \approx 63$  nuqta) bilan bog'lab natijani izohlang. *Ko'rsatma:* oyna sekin komponentning ( $0,3t$ , davri  $\approx 209$  nuqta) kamida bir qismini qamrashi kerak.

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